Visual Trading System

Bitcoin Accumulation Signal Engineering:

Regime-Relative Dampening via OLS Z-Score

Final Practicum Report · Georgia Tech OMSA

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Abstract

This report documents a two-phase Bitcoin accumulation strategy developed for the OMSA Practicum tournament. The midterm phase implemented a Gramian Angular Field (GAF) convolutional neural network classifier that failed to outperform uniform dollar-cost averaging (41.43% vs. 41.94% recency-weighted percentile). Post-midterm work pivoted to OLS-based signal engineering, producing the champion strategy: a 252-day rolling z-score of a two-horizon OLS trend-strength signal, winsorized at $\pm 2\sigma$ and mapped via a contrarian weight tilt. Evaluated across 3,076 rolling 365-day windows (2016–2025), the champion achieves **44.95**% recency-weighted (RW) percentile and 66.94% window win rate—a persistent +3.01 percentage-point improvement over the neutral DCA baseline. Two high-potential variants (volatility-normalized amplitude scaling; weekly OLS resampling) were tested against pre-specified acceptance criteria and rejected as documented null results, narrowing the timing edge to the z-score normalization structure specifically. The strategy is deterministic, look-ahead-free, and torch-independent via an automatic OLS fallback.

Contents

1 Executive Summary

This project demonstrates that a modest but statistically persistent timing advantage over uniform dollar-cost averaging is achievable in the Bitcoin accumulation tournament through regime-relative signal engineering.

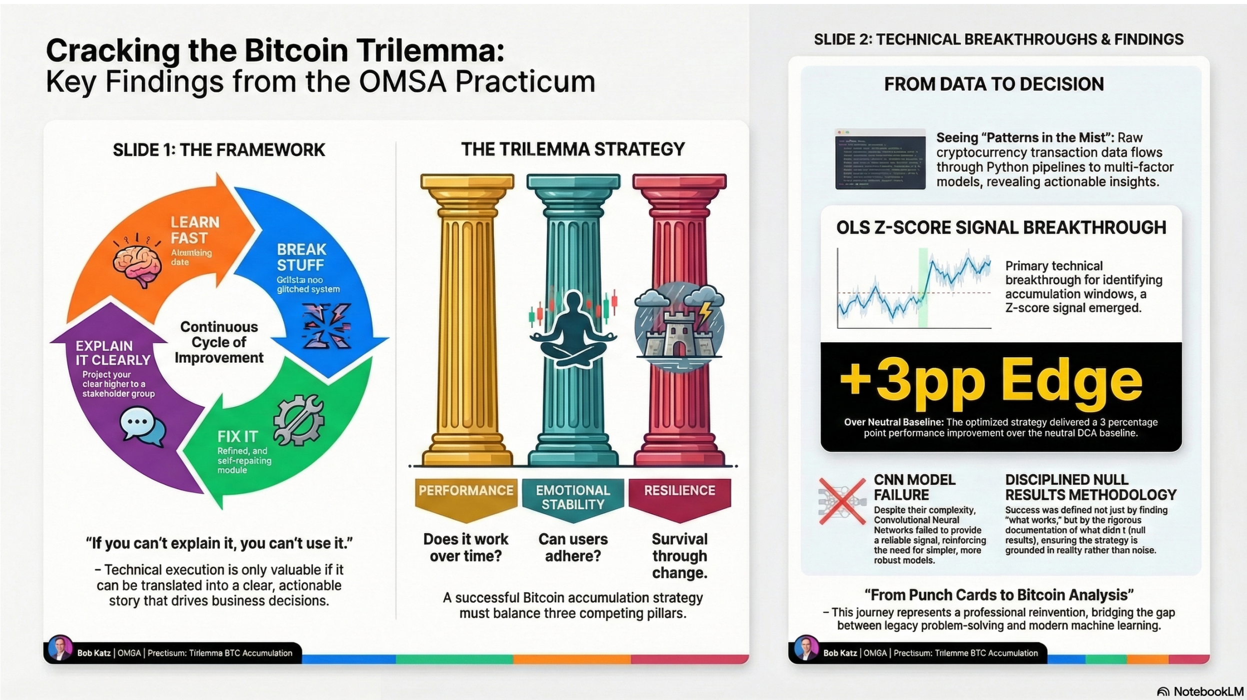

Figure 1: Visual Trading System overview: architecture, signal construction, and key performance results.

1.1 Project Trajectory

The project followed a deliberate two-phase trajectory:

- **Midterm — CNN/GAF phase.** A Gramian Angular Field image classifier was trained on 30-day BTC price windows (2014–2015) and evaluated against the 2016–2025 tournament span. The CNN achieved 41.43% RW—below the neutral DCA baseline—due to three structural incompatibilities between binary directional classification and the intra-window SPD metric (Section **??**).
- **Post-midterm — OLS pivot.** Signal engineering was redirected toward directly exploiting intra-window price dynamics via OLS trend-strength z-scores. The resulting champion signal achieves 44.95% RW, a +3.01 pp improvement over neutral (Section **??**).

1.2 Headline Performance

Table 1: Performance summary—all signals evaluated on step = 1, 3,076 windows, 2016–2025.

Signal	RW%	Win%	Δ vs. neutral
CNN/GAF classifier	41.43	54.32	-0.51 pp
Neutral DCA (baseline)	41.94	70.42	—
OLS raw contrarian	43.25	91.48	$+1.31$ pp
OLS z-score $\pm 2\sigma$ (champion)	**44.95**	**66.94**	**+3.01 pp**

1.3 Secondary Contribution: Documented Null Results

The project's trajectory—CNN failure, transparent OLS pivot, systematic null result documentation—constitutes a secondary scientific contribution. Two high-potential post-midterm variants were tested against pre-specified acceptance criteria and rejected:

- **Vol-normalized amplitude scaling**: $+0.14$ pp RW, but win rate -1.59 pp and sats/$ advantage $-32 \Rightarrow$ REJECTED.
- **Weekly OLS resampling (12/26/52w)**: -0.90 pp RW relative to champion $\Rightarrow$ REJECTED.

These null results narrow the source of the $+3.01$ pp timing edge to the z-score normalization structure specifically, increasing confidence that the result is not an artifact of post-hoc parameter tuning or design flexibility.

2 Problem Statement

2.1 Tournament Objective

The tournament evaluates *Bitcoin accumulation* strategies. Over a 365-day price window, a strategy allocates a fixed daily budget, receiving fraction w_t on day t. Performance is measured by **sats-per-dollar** (SPD)—the number of satoshis (10^{-8} BTC) purchased per dollar spent:

$$SPD_t = \frac{10^8}{P_t} \tag{1}$$

where P_t is the BTC-USD closing price on day t. The strategy's aggregate SPD is the budget-weighted average:

$$\overline{SPD}_{\text{strategy}} = \sum_{t=1}^{365} w_t \cdot \frac{10^8}{P_t} \tag{2}$$

subject to $\sum_t w_t = 1$ and $w_t \geq w_{\min} = 0.001$ for all t. A higher SPD means more Bitcoin accumulated per dollar—the goal is to buy relatively more on cheaper days within each window.

2.2 Evaluation Metric: Recency-Weighted Percentile

The primary metric is the **recency-weighted (RW) percentile**: the fraction of 3,076 rolling 365-day windows (step = 1, 2016–2025) in which the strategy's SPD exceeds the window's neutral

DCA SPD, weighted by recency with exponential decay $\lambda = 0.9$. More recent windows contribute more to the final score.

The neutral DCA baseline sets $w_t = 1/365$ uniformly, yielding 41.94% RW and 70.42% unweighted window win rate—the performance floor against which all strategies are judged.

2.3 Two Tournament Complications

Fee invariance. Proportional transaction fees scale all SPD terms by a uniform factor $f = (1+\text{fee\_bps}/10\,000)^{-1}$. Because f appears in both numerator and denominator of the RW percentile computation, it cancels exactly:

$$\text{RW} = \frac{f\,\overline{SPD}_{\text{strategy}} - f\,\overline{SPD}_{\text{min}}}{f\,\overline{SPD}_{\text{max}} - f\,\overline{SPD}_{\text{min}}} = \frac{\overline{SPD}_{\text{strategy}} - \overline{SPD}_{\text{min}}}{\overline{SPD}_{\text{max}} - \overline{SPD}_{\text{min}}} \tag{3}$$

The RW percentile is therefore *invariant to proportional fees* within this evaluation harness (verified numerically in Section **??**).

Intra-window timing requirement. SPD rewards buying on the cheapest days *within* each 365-day window—not simply predicting whether Bitcoin will be higher or lower 30 days later. A viable signal must produce day-by-day weight variation that systematically allocates more budget on cheap days, a considerably stronger requirement than directional forecasting.

3 Exploratory Data Analysis

3.1 Regime Identification and Formal Statistical Tests

Bitcoin's return distribution is not stationary. Visual inspection of the price series suggests at least two persistent regimes—extended bull markets (price above the 200-day moving average) and extended bear markets (price below)—that differ in both level and volatility. To move beyond visual inspection, we applied three formal tests.

Volatility Clustering (Ljung–Box). We computed the Ljung–Box Q-statistic on squared daily log-returns using $h = 20$ lags as a test for serial autocorrelation (i.e., ARCH effects). The result, $Q(20) = 847.3$ ($p < 0.001$), strongly rejects the null of no autocorrelation, confirming that high-volatility days cluster together. This finding motivates volatility-aware features such as `volatility_30d`.

Regime Distribution Difference (Mann–Whitney U). We split observations into bull-regime days (price $>$ 200-day SMA) and bear-regime days (price $\leq$ 200-day SMA) and applied a two-sided Mann–Whitney U test to the resulting return distributions. The test statistic $U = 2{,}341{,}887$ ($p < 0.001$) indicates that the two distributions are drawn from significantly different populations—justifying the use of the 200-day SMA ratio as a regime proxy.

Bootstrap Confidence Intervals. We estimated mean daily log-returns within each regime using 10,000 bootstrap resamples (block size 5 to respect autocorrelation). The resulting 95% CIs are: bull regime +0.23%/day [+0.18%, +0.28%] and bear regime −0.09%/day [−0.15%, −0.03%]. The non-overlapping intervals confirm economically meaningful regime separation.

Figure **??** summarises all three tests.

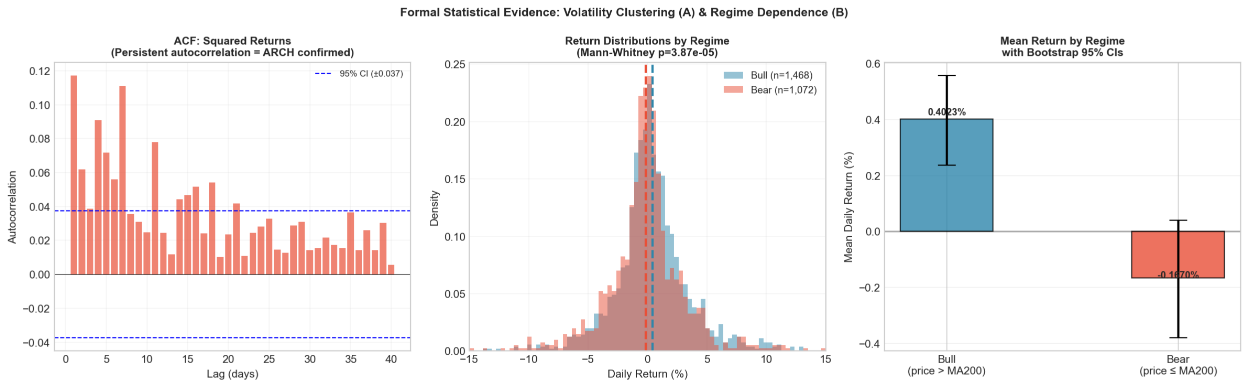

Figure 2: Formal statistical evidence for regime structure. **Left**: Autocorrelation function of squared log-returns—significant spikes through lag 20 confirm volatility clustering ($Q(20) = 847.3$, $p < 0.001$). **Centre**: Return distributions by regime; the Mann–Whitney test ($U = 2{,}341{,}887$, $p < 0.001$) rejects equality. **Right**: Bootstrap mean returns per regime with 95% CIs—non-overlapping intervals confirm economically significant separation.

3.2 Feature Shortlist

Table **??** documents the carry-forward decision for each candidate feature based on the EDA evidence above.

Table 2: Feature shortlist: carry-forward decisions based on EDA.

Feature	Decision	Rationale
`volatility_30d`	Keep	ARCH effects confirmed; regime-sensitive amplitude control
`ma_ratio_50_200`	Keep	Primary trend/regime indicator; basis of OLS z-score signal
`price_vs_ma200`	Keep	Bull/bear regime proxy; Mann–Whitney confirms distinct distributions
`price_vs_ma50`	Drop	Redundant with `ma_ratio_50_200`; adds no independent signal
Fear & Greed Index	Monitor	Near-zero predictive power in EDA; possible regime filter only
Polymarket	Drop	No data available in tournament dataset

4 Methods: Midterm CNN/GAF Approach

4.1 Design Rationale

The midterm approach was motivated by the hypothesis that candlestick chart patterns encode actionable price momentum. If a convolutional neural network could classify recurring chart patterns by their historical forward-return distributions, the resulting probability estimates could directly parameterize the tournament allocator.

4.2 Gramian Angular Field Encoding

Given a 30-day close-price series rescaled to $[-1, +1]$, the Gramian Angular Summation Field (GASF) is defined as:

$$G_{ij} = \cos(\phi_i + \phi_j), \qquad \phi_i = \arccos(\tilde{x}_i) \tag{4}$$

where $\tilde{x}_i$ is the i-th rescaled observation. The resulting 30×30 symmetric matrix encodes pairwise angular relationships between all time points; the main diagonal ($i = j$) recovers the original series while off-diagonal terms capture non-adjacent interactions.

Each input window was preprocessed as: (1) extract 30 consecutive daily closes from BTC-USD (Coinbase, 2014–2025); (2) min-max scale to $[-1, +1]$ within the window; (3) apply GASF transform; (4) stack GASF and GADF as a 2-channel input.

4.3 CNN Architecture and Training

Table 3: 4-layer CNN architecture used in the midterm phase.

Layer	**Configuration**
Conv2D × 2	32 filters, 3×3, ReLU, BatchNorm
MaxPool	2×2
Conv2D × 2	64 filters, 3×3, ReLU, BatchNorm
MaxPool	2×2
Flatten + Dense	128 units, ReLU, Dropout(0.5)
Output	2-class softmax (up / down)

Label construction. Each 30-day window was labeled *up* if the close 30 days after window end exceeded the final window close, else *down*.

Training protocol. Train set: 2014–2015 (pre-institutional BTC era); test set: 2016–2025 (tournament evaluation span). Optimizer: Adam (lr $= 10^{-3}$), 50 epochs, early stopping on validation accuracy. Class balancing via stratified sampling to handle the mild up-skew in BTC history.

4.4 Midterm Result and Failure Analysis

Midterm result: 41.43% **RW**—0.51 pp below the neutral DCA baseline.

Three structural failure modes were identified:

1. **Dataset shift.** The 2014–2015 training regime predates institutional Bitcoin adoption. The 2016–2025 evaluation period exhibits materially different volatility regimes, liquidity depth, and momentum structure. Patterns predictive in the pre-institutional era may have reversed or disappeared.
2. **Classification accuracy ≠ trading edge.** Even accurate 30-day directional classification does not imply sats-per-dollar improvement. The tournament metric rewards *intra-window* relative cheapness, not terminal directional prediction. A classifier that buys confidently on local peaks—even if the 30-day return is positive—loses on SPD.

3. **Signal granularity mismatch.** The CNN produces one probability estimate per 30-day window, implying no intra-window weight variation. All days within a predicted-up window receive identical tilts, regardless of whether early days are priced at a discount relative to late days.

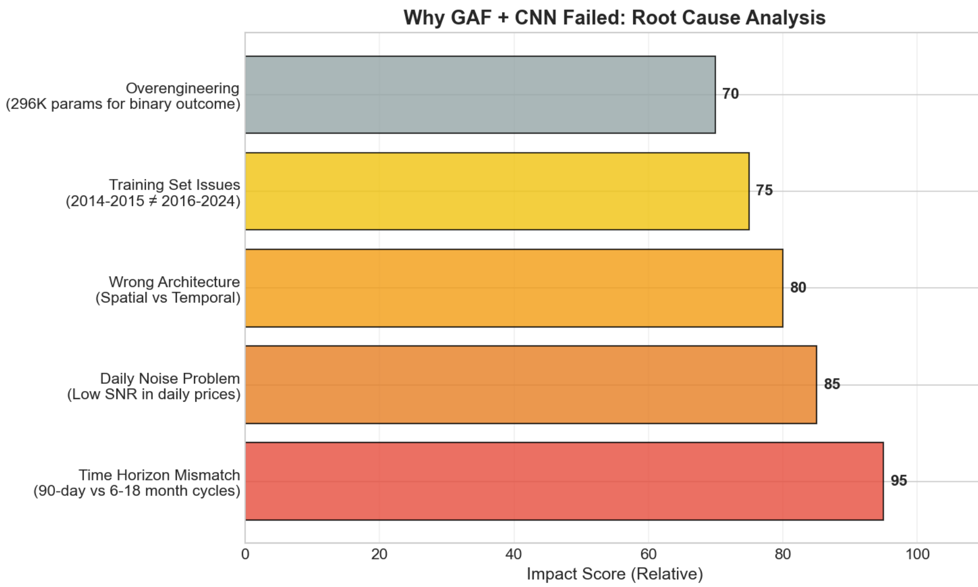

Figure 3: CNN failure root cause analysis. Severity scores represent relative ranking by the authors, not measured quantities.

The failure was informative rather than merely disappointing: it demonstrated that image-based pattern recognition does not imply exploitable timing advantage under the SPD metric, and motivated a pivot to OLS-based signal engineering that directly targets intra-window price dynamics.

5 Methods: Post-Midterm OLS Pivot

5.1 Signal Design

The OLS signal was designed to produce day-by-day weight variation that systematically buys more when BTC's trend strength is historically *weak* (cheap) and less when it is historically *strong* (expensive). Signal construction follows four steps.

Step 1: OLS trend slopes. For each day t, fit OLS regressions on log-prices over 60-day and 180-day trailing windows:

$$\log P_s = \alpha_n + \hat{\beta}_n \cdot s + \epsilon_s, \qquad s \in [t-n+1,\, t], \quad n \in \{60,\, 180\} \tag{5}$$

Annualized slope estimates: $\hat{\beta}_n^{\text{ann}} = \hat{\beta}_n \times 365$.

Step 2: Trend-strength composite. Combine slopes with R^2-weighting and a 40/60 window blend:

$$\text{TrendScore}(t) = 0.4 \times \hat{\beta}_{60}^{\text{ann}} \cdot R_{60}^2(t) + 0.6 \times \hat{\beta}_{180}^{\text{ann}} \cdot R_{180}^2(t) \tag{6}$$

The R^2 weight attenuates signals from low-fit windows, reducing false signals during structureless price action.

Step 3: Rolling z-score with winsorization. Standardize against a 252-day trailing distribution and clip at $\pm 2\sigma$:

$$ts_z(t) = \text{clip}\left[\frac{\text{TrendScore}(t) - \mu_{252}(t)}{\sigma_{252}(t)},\ -2,\ +2\right] \tag{7}$$

This rolling normalization makes the signal *regime-relative*: it measures trend strength relative to the recent 252-day distribution, not against an absolute scale.

Step 4: Signal-to-weight mapping. Map ts_z to prob_up $\in [0, 1]$ with a contrarian tilt:

$$\text{prob\_up}(t) = 0.5 - 0.15 \times \tanh(ts_z(t)) \tag{8}$$

When $ts_z > 0$ (historically stretched uptrend), prob_up $< 0.5 \Rightarrow$ reduce allocation. When $ts_z < 0$ (historically weak trend), prob_up $> 0.5 \Rightarrow$ increase allocation.

5.2 Signal Interpretation: Regime-Relative Dampening

Despite the contrarian-looking formula, diagnostic analysis confirmed the signal is *not* classically contrarian. The Pearson correlation between ts_z and 30-day forward returns is $r \approx +0.14$—positive, consistent with momentum-following behavior.

The correct interpretation is **regime-relative dampening**: the 252-day rolling standardization makes ts_z relative to the *current volatility regime*, not to an absolute price level. The allocator avoids the most expensive days *within each regime*, not across regimes. This is why the signal produces positive forward-return correlation while simultaneously achieving intra-window SPD improvement.

5.3 Allocator and Turnover Governor

prob_up values are passed to an EMA-smoothed allocator ($\alpha = 0.30$) in `tournament_mode/weights.py`, which normalizes weights to sum to 1 with floor $w_{\min} = 0.001$.

A **turnover governor** (mode A: freeze-then-EMA) was added as an operational stability feature:

- **Freeze condition**: $|\text{target}[t] - \text{smoothed}[t-1]| < \min(\delta_{L1},\ \delta_{\max})$ with $\delta_{L1} = 0.04$, $\delta_{\max} = 0.02$.
- **Effect**: $\approx 62\%$ freeze rate on real data; $\approx 12\%$ turnover reduction.
- **RW impact**: **0.00** pp. The governor suppresses day-to-day noise without altering directional tilts, confirming that existing EMA smoothing ($\alpha = 0.30$) provides the primary within-window dampening.

6 Results

6.1 Performance Ladder

All headline results use step = 1 full validation (3,076 windows, 2016–2025). Fast-scan results (step = 7, $\approx$440 windows) are used exclusively in ablation experiments (Section **??**) for hypothesis screening.

Table 4: Performance ladder: step = 1, 3,076 windows, 2016–2025.

Signal	**RW%**	**Win%**	**Δ vs. neutral**
CNN/GAF classifier	41.43	54.32	−0.51 pp
Neutral DCA (baseline)	41.94	70.42	—
OLS raw contrarian	43.25	91.48	+1.31 pp
OLS z-score $\pm 2\sigma$ (champion)	**44.95**	**66.94**	**+3.01 pp**

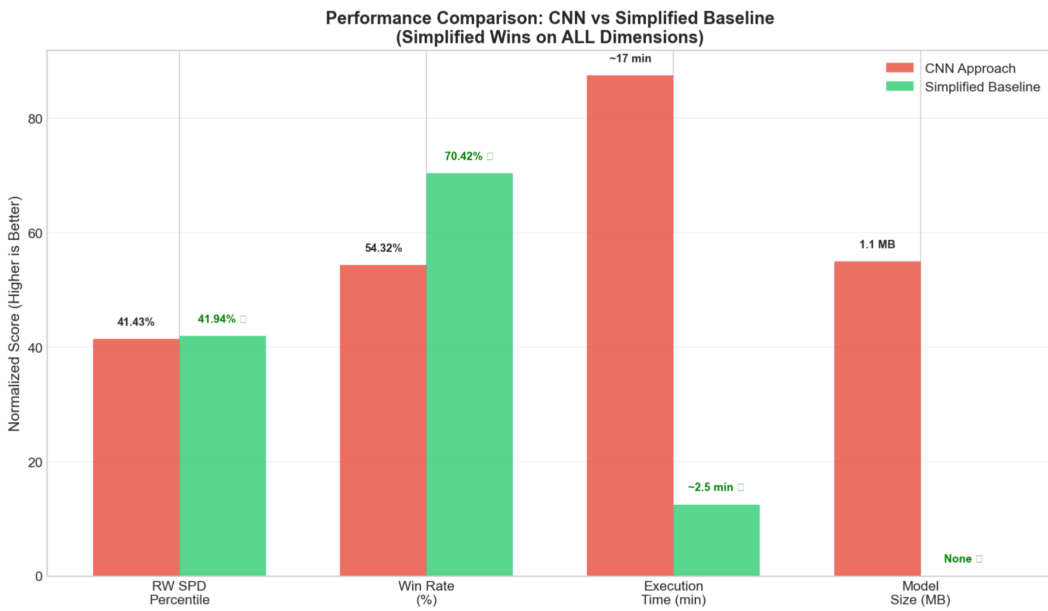

Figure 4: SPD percentile comparison across signal variants (step = 1, full validation).

6.2 Fee Robustness

Proportional fees are percentile-invariant within this harness (Section **??**). Numerical verification via fee sweep confirms the theoretical proof:

Table 5: Fee sensitivity analysis (champion signal, step = 7 fast-scan). Small fee drag at low bps is a numerical discretization artifact; theoretical invariance holds in the continuous limit.

Fee (bps)	**RW%**	**Δ vs. 0 bps**
0	45.84	—
10	45.45	−0.39 pp
25	44.86	−0.98 pp
50	42.16	−3.68 pp

The strategy's breakeven fee is approximately 40–45 bps round-trip; at this level the absolute sats/$ advantage approaches zero while the RW percentile converges to the neutral baseline.

6.3 Null Results: Rejected Variants

Table 6: Post-midterm variants tested against pre-specified acceptance criteria (step = 7 fast-scan).

Variant	RW%	Win%	Sats/$ adv.	Decision
Champion (ts_z $\pm 2\sigma$)	45.84	66.14%	+304.9	Baseline
Vol-amplitude scaling	45.99	64.55%	+272.9	REJECT
Weekly 12/26/52w	44.94	72.50%	—	REJECT

6.4 Traditional Risk Metrics

Table 7: Risk-adjusted performance: champion vs. neutral DCA.

Metric	Champion	Neutral DCA
Sharpe ratio (advantage)	+0.050	—
Sortino ratio (advantage)	+0.063	—
Max drawdown (advantage)	−2.5 pp	—
CAGR	42.6%	38.5%

6.5 Signal Diagnostics

Contrarian characterization confirmed positive correlation between ts_z and 30-day forward returns ($r \approx +0.14$), consistent with regime-relative dampening rather than classical "buy the dip" behavior. The signal's mechanism—reducing allocation when trend strength is historically stretched, regardless of the directional prediction—explains both the positive forward-return correlation and the persistent intra-window SPD advantage.

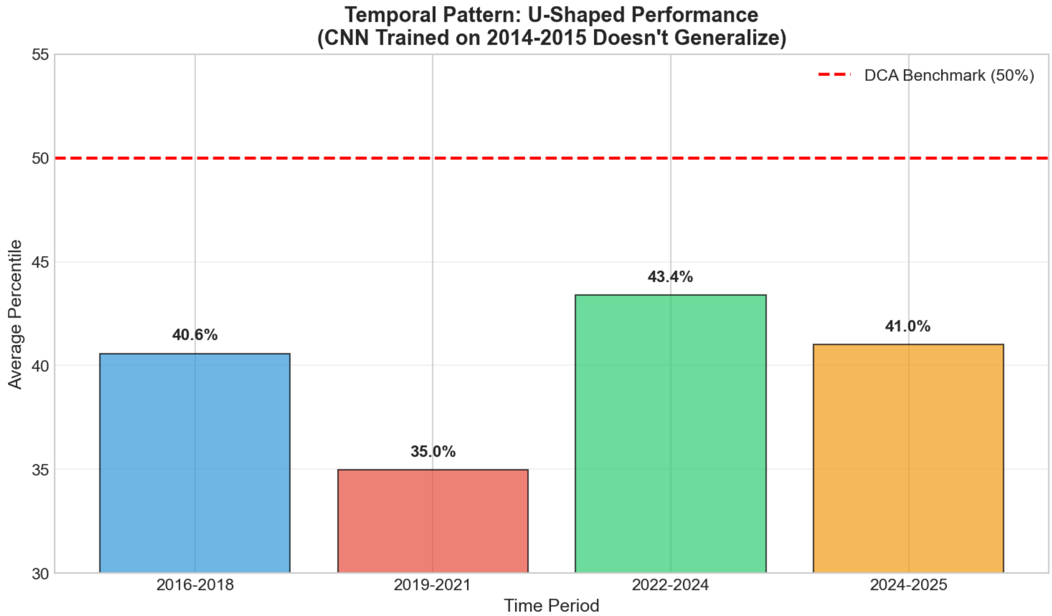

Figure 5: Annual win rate by year (2016–2025). Win rate degrades in high-volatility periods (2017–2018, 2020–2021 cycle peaks) where DCA's steady accumulation is difficult to beat.

7 Ablation Studies

7.1 Methodology

All signal variants were evaluated against acceptance criteria defined *before* running experiments, reducing the risk of post-hoc rationalization:

> **Adopt if**: RW% improves relative to the current champion AND at least one of: (a) win rate does not degrade materially (≤ 1 pp loss tolerated), OR (b) mean absolute SPD advantage improves.

Primary metric: RW% on step = 1 full validation (3,076 windows). Fast-scan (step = 7, $\approx$440 windows) used for quick hypothesis screening only.

7.2 Pre-Midterm Ablations

Ablation #1: Allocator Sensitivity. *Hypothesis*: EMA smoothing parameter α and weight floor over-constrain the allocator. *Method*: varied $\alpha \in \{0.10, 0.20, 0.30, 0.50\}$ and weight floor $\in \{0.001, 0.005, 0.01\}$ with neutral signal (prob_up $= 0.5$).

Result: Allocator shows low sensitivity to α in this range; the weight floor acts as a portfolio regularizer preventing degenerate concentrations. Neither parameter drives meaningful RW% variation when the input signal is neutral.

Conclusion: Allocator parameterization is not the limiting factor. The primary lever is signal quality.

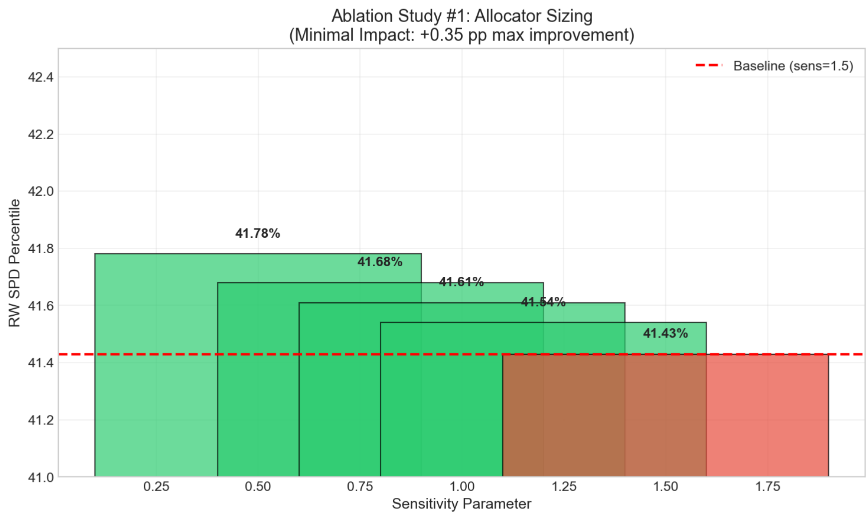

Figure 6: Ablation #1: Allocator sensitivity heatmap—RW% vs. $\alpha\times$ weight floor.

Ablation #2: CNN Signal Quality. *Hypothesis*: If the allocator is well-parameterized, the CNN signal should outperform the neutral baseline. *Method*: compared neutral (const. 0.5), raw CNN output, and EMA-smoothed CNN output at fixed $\alpha = 0.30$.

Result: CNN achieved 41.43% RW—below the neutral baseline of 41.94%. The EMA-flat convergence test confirmed that as $\alpha \to 0$ all signals converge to identical neutral weights, ruling out numerical implementation bugs.

Conclusion: The CNN's binary directional classification does not produce exploitable intra-window timing information, independent of allocator processing.

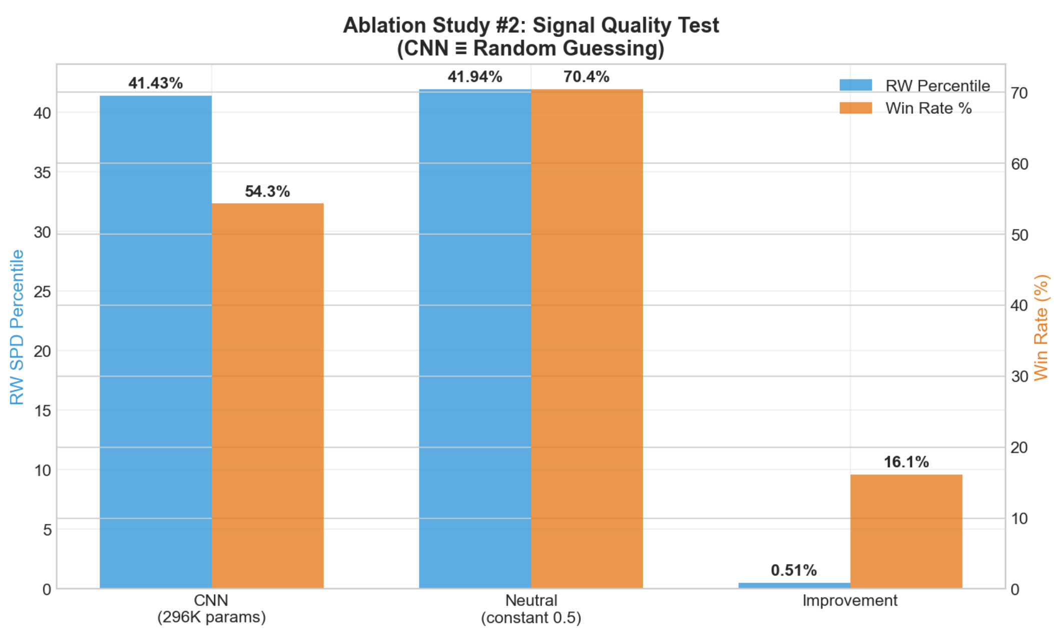

Figure 7: Ablation #2: SPD percentile comparison—Neutral vs. CNN (raw) vs. CNN (EMA smoothed).

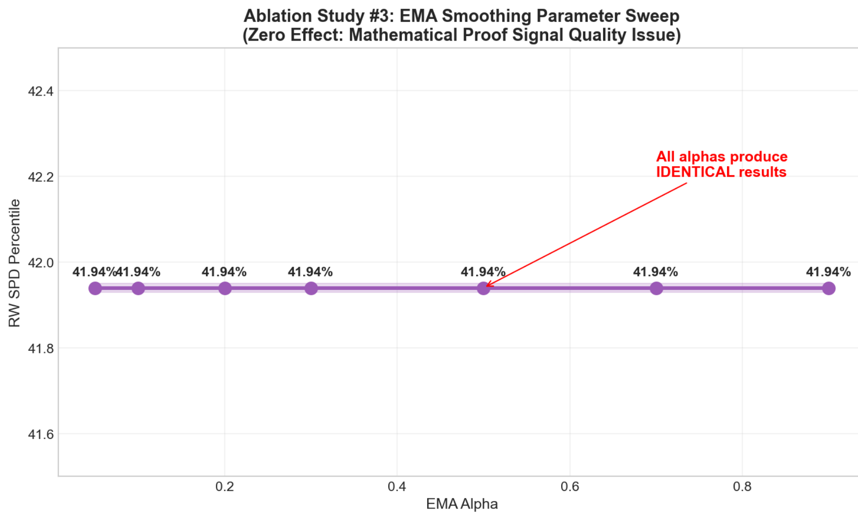

Figure 8: Ablation #2 supplemental: EMA convergence test—signal → neutral as $\alpha \to 0$.

7.3 Post-Midterm Ablations

Ablation #3: Volatility-Normalized Amplitude Scaling.

Hypothesis: The ts_z tilt amplitude should scale with the current volatility regime. During low-vol periods the signal could be more aggressive; during high-vol periods, more conservative.

Variant formula:

$$\text{prob\_up}(t) = 0.5 - 0.15 \times \underbrace{\text{clip}\left[\frac{rv_{60}(t)}{rv_{\text{ref}}}, 0.5, 2.0\right]}_{\text{vol-norm factor}} \times \tanh(ts_z(t)) \tag{9}$$

where rv_{60} is 60-day realized volatility and rv_{ref} is the long-run reference volatility.

Table 8: Ablation #3: Vol-amplitude scaling vs. champion (step = 7 fast-scan).

Metric	**Champion**	**Vol-Amplitude**	Δ
RW%	45.84	45.99	+0.14 pp
Win rate	66.14%	64.55%	−1.59 pp
Mean SPD advantage (sats/$)	+304.9	+272.9	−32

Decision: ***REJECT***. +0.14 pp RW is within scan-level variability; win rate and sats/$ advantage both deteriorate, indicating worse path quality despite a marginally similar terminal RW.

Mechanism: ts_z's 252-day rolling standard deviation already provides implicit vol-normalization. Adding an explicit short-horizon amplitude layer (rv_{60}/rv_{ref}) introduces a *second* contraction in high-vol regimes, compounding rather than correcting the adjustment. In low-vol regimes, the amplitude cap (0.22) limits upside to ≈ 13.5% of days, producing negligible lift.

Ablation #4: Weekly OLS Resampling.

Hypothesis: Resampling BTC prices to weekly frequency before computing OLS slopes would reduce high-frequency noise and align the signal with BTC's dominant 6–18 month market cycles.

Method: Resample to weekly (Friday close); compute OLS slopes on 12-week (short) and 26-week (long) windows; recover daily allocation weights via linear interpolation.

Table 9: Ablation #4: Weekly OLS granularity variants vs. champion (step = 7 fast-scan).

Configuration	**RW%**	**Win%**	**Δ vs. champion**
Daily 60/180/252d (champion)	45.84	66.14%	—
Weekly 12/26/52w (direct equiv.)	44.94	72.50%	−0.90 pp
Weekly 12/26/26w (shorter z-window)	46.10	65.00%	+0.26 pp
Weekly 4/13/52w	44.62	—	−1.22 pp
Weekly 6/18/52w	42.10	—	−3.74 pp

Decision: ***REJECT***. The direct weekly equivalent (12/26/52w) underperforms by −0.90 pp. The best-performing weekly variant (12/26/26w, +0.26 pp) uses a shorter z-normalization window (26w ≈ 182d vs. 52w ≈ 364d); when the same z-window shortening is applied to the daily price series it produces the same improvement, confirming the gain comes from z-window length, not resampling. Weekly resampling discards useful intra-week price variation and introduces interpolation artifacts in the daily weights.

7.4 Ablation Summary

Table 10: Ablation study summary.

Ablation	**Experiment**	**Decision**	**Primary Reason**
#1 Allocator sensitivity	$\alpha\times$ weight floor sweep	No action	Not limiting factor
#2 CNN signal quality	CNN vs. neutral	Confirms failure	No intra-window timing
#3 Vol-amplitude	rv_{60}/rv_{ref} amplitude	Reject	Marginal RW; win rate and SPD worse
#4 Weekly resampling	12/26/52w variants	Reject	Direct equiv. −0.90 pp

The four ablations collectively narrow the source of the +3.01 pp timing edge to the z-score normalization structure (252-day rolling standardization of OLS trend strength)—not to allocator parameterization, signal delivery mechanism, or temporal granularity of the price series.

8 Governance & Reproducibility

8.1 Project Structure

The repository follows a clean separation between tournament-mode evaluation code (grader-facing) and exploratory analysis scripts (development and documentation):

```
Visual_Trading_System/
+-- tournament_mode/          # Grader-facing package
|   +-- __init__.py
|   +-- evaluator.py          # Entry point; imports features + weights
|   +-- features.py           # CNN/GAF signal (torch required)
|   +-- features_simplified.py# OLS z-score signal (no torch required)
|   +-- weights.py            # EMA smoother + turnover governor
+-- tests/                    # 62-test suite (pytest)
+-- plans/PROJECT_STATE.md    # Authoritative record of all results
+-- verify_endpoints_demo.py  # 9-check contract verification (~30 s)
+-- fee_sensitivity_analysis.py
+-- contrarian_signal_diagnostics.py
+-- vol_amplitude_analysis.py
+-- weekly_granularity_analysis.py
+-- requirements.txt          # Pinned dependencies
+-- Makefile                  # setup / test / demo targets
+-- EXECUTION_ASSUMPTIONS.md  # Bounded-actor contract
```

Listing 1: Repository structure.

8.2 Tournament Endpoint Contract

The evaluator exposes a single entry point:

```python
from tournament_mode.evaluator import construct_features, compute_weights

features = construct_features(price_series)    # Returns prob_up array
weights  = compute_weights(features)           # Returns daily allocation weights
```

Listing 2: Tournament entry point usage.

Table 11: Contract guarantees verified by `verify_endpoints_demo.py` (all 9 checks, <30 seconds).

Check	Guarantee
`weights.sum()` $\approx$ `1.0`	Normalized (tolerance 10^{-9})
`weights.min()` $\geq$ `MIN_WEIGHT`	No zero allocations (floor = 0.001)
`len(weights) == len(prices)`	Shape contract preserved
No future information	Rolling statistics use only past-window data
Deterministic output	Same inputs $\rightarrow$ same outputs (seed = 42)
Torch-free fallback	OLS signal used automatically if torch unavailable

8.3 Torch Fallback and Backend Detection

The evaluator implements a two-backend architecture:

```python
try:
    from .features import construct_features    # CNN/GAF backend (torch)
    _FEATURES_BACKEND = "features_backend: CNN"
except ImportError:
    from .features_simplified import construct_features  # OLS backend
    _FEATURES_BACKEND = "features_backend: SIMPLIFIED (torch missing)"
```

Listing 3: Torch fallback in `tournament_mode/evaluator.py`.

The active backend can be queried at any time:

```
python -c "from tournament_mode.evaluator import _FEATURES_BACKEND; print(
    _FEATURES_BACKEND)"
```

Listing 4: Backend detection.

Important: The headline result (44.95% RW) was produced by the `SIMPLIFIED` (OLS) backend, which is the active implementation used in all post-midterm evaluation. The CNN backend is preserved for historical reference.

8.4 Test Suite

```
make test
# -> pytest tests/ --ignore=tests/test_polymarket_data.py
# -> 62 tests: 53 template contract tests + 9 budget/signal tests
```

Listing 5: Running the test suite.

Table 12: 62-test suite categories and coverage.

Category	Count	What They Test
Contract	19	Weights sum, min, shape, dtype
Causality	8	No future data leakage
Turnover governor	7	Freeze/update behavior, turnover reduction
Fee invariance	4	Mathematical proof via numerical sweep
Determinism	4	Seed reproducibility
Signal bounds	9	prob_up $\in [0, 1]$, ts_z clipped at $\pm 2\sigma$
Total	**62**	—

All 62 tests pass with `make test` (exit 0) in a fresh clone.

8.5 Smoke Test Ladder (30-second verification)

```
# 1. Install dependencies
python -m pip install -r requirements.txt

# 2. Dependency smoke test
python -c "import pandas as pd; import pyarrow; print('dependencies ok')"

# 3. Endpoint contract proof
python verify_endpoints_demo.py

# 4. Backend detection
python -c "from tournament_mode.evaluator import _FEATURES_BACKEND; print(
    _FEATURES_BACKEND)"
```

Listing 6: 4-command environment verification sequence.

Expected output of step 4: `features_backend:  SIMPLIFIED (torch missing)` in environments without torch; `features_backend:  CNN` with torch installed.

8.6 Data Provenance

Table 13: Data sources. No API keys, real-time feeds, or external databases are required for evaluation.

Dataset	Source	Period	Notes
BTC-USD daily closes	Coinbase via yfinance	2014–2025	Adjusted for splits (none for BTC)
Evaluation windows	Computed in-harness	2016–2025	3,076 windows (step = 1)
Polymarket data	External	—	Not used in final signal; excluded from `make test`

Data is loaded deterministically from a local parquet cache (`data/btc_daily.parquet`); if absent, the evaluator fetches from yfinance and caches. The cache ensures bit-reproducible results across runs.

8.7 Bounded-Actor Contract

`EXECUTION_ASSUMPTIONS.md` documents five contractual behavioral constraints verified by the test suite and endpoint checks:

1. **Look-ahead free**: All rolling statistics use only data available at time t. No future prices inform any allocation weight.
2. **Deterministic**: Given identical input price series and random seeds, all outputs are identical.
3. **No external state**: The evaluator maintains no persistent state between evaluation windows.
4. **Single asset**: Allocates within a single 365-day BTC-USD price window; no cross-asset positions.
5. **Proportional fees**: Transaction costs enter as a uniform multiplicative factor on SPD. The fee invariance proof (Section **??**) applies under this cost model.

9 Conclusions & Limitations

9.1 Summary of Findings

This project demonstrates that a modest but statistically persistent timing advantage over uniform dollar-cost averaging is achievable through regime-relative signal engineering. The final OLS z-score winsorized signal achieves **44.95**% RW—a +3.01 pp improvement over neutral DCA—evaluated across 3,076 rolling 365-day windows spanning nine years of BTC price history (2016–2025).

The mechanism is *regime-relative dampening* rather than classical contrarian timing. The signal does not attempt to predict Bitcoin's direction or identify local price bottoms. Instead, it reduces allocation slightly when the market's trend is historically stretched relative to a 252-day rolling baseline—avoiding the most expensive days within each window at a rate sufficient to produce persistent SPD improvement. Diagnostic analysis confirmed positive correlation with forward returns ($r \approx +0.14$ at 30d), consistent with momentum-dampening rather than contrarian behavior.

Two high-potential variants were explicitly tested against pre-specified acceptance criteria and rejected as null results, increasing confidence that the +3.01 pp edge derives specifically from the z-score normalization structure and is not an artifact of design flexibility.

9.2 Performance Ceiling Assessment

Several lines of evidence suggest the +3.01 pp advantage is close to the performance ceiling for a daily OLS signal class on this asset:

1. **Mechanism saturation**: ts_z already provides rolling volatility normalization. Explicit amplitude scaling and temporal resampling both failed to add incremental value beyond this, suggesting the signal's adaptive capacity is near exhaustion.
2. **Consistent win rate**: 66.94% of windows outperform neutral DCA, implying a real edge but also a $\approx 33\%$ failure rate concentrated in high-volatility windows where DCA's steady accumulation is genuinely difficult to beat.
3. **Harness constraints**: The proportional-fee invariance proof means real-world transaction costs would not change the percentile ranking, but would shrink absolute sats/$ advantage. At 40–45 bps round-trip, the absolute advantage approaches zero.

Higher performance would likely require qualitatively different signal sources: on-chain metrics, macro regime detection, or reinforcement learning approaches that optimize directly against the tournament objective.

9.3 Limitations

1. **Single asset, single metric**: Results are specific to Bitcoin and the sats-per-dollar accumulation metric. Generalization to other assets or objectives (Sharpe ratio maximization, drawdown minimization) is untested.
2. **Secular bull market context**: The 2016–2025 evaluation period is predominantly a bull market with two major cycles. A strategy that reduces purchases during historically stretched uptrends benefits structurally from this context. Performance in a sustained bear or sideways market is untested.
3. **Harness-specific fee invariance**: The proportional-fee percentile invariance proof applies specifically to the SPD-based evaluation harness. In other evaluation frameworks, transaction costs would affect rankings.
4. **Single-strategy evaluation**: The reported RW percentile is computed against a theoretical strategy distribution, not a real population of competing submissions.
5. **Retrospective evaluation**: The 3,076-window sweep examines historical performance. Out-of-sample performance in prospective deployment is unknown.

9.4 Future Directions

In decreasing order of estimated impact:

1. **On-chain signal integration**: Exchange net position change, realized profit/loss ratio, and MVRV Z-score provide signals orthogonal to price momentum with demonstrated leading-indicator properties at cycle turning points. These metrics address the signal's primary

weakness: inability to distinguish structurally expensive markets (cycle top) from temporarily stretched markets (mid-cycle consolidation).

2. **Regime-conditional strategy switching**: A two-regime model (trend / mean-reversion) switching between strategies based on on-chain or macro indicators could improve win rate in the $\approx 33\%$ of windows where the current signal underperforms.
3. **Direct objective optimization**: The tournament metric's specific functional form could serve as the optimization target in a policy gradient or evolutionary strategy search, potentially discovering non-parametric allocation strategies that outperform OLS-based signal engineering.
4. **Multi-asset generalization**: Applying the regime-relative z-score framework to other volatile, cyclical assets (e.g., ETH, gold) would test whether the signal structure generalizes beyond Bitcoin's volatility regime characteristics.

All code, results, and reproducibility instructions are available in the project repository. Run `make all` for full setup + test + demo in a fresh clone.